(v) $\cos 20^\circ - \cos 30^\circ$ (vi) $\cos 75^\circ + \cos 45^\circ$

(vii) $\cos 55^\circ + \sin 55^\circ$

Solution

(i) $\sin 9A + \sin 7A = 2\sin\left(\frac{9A+7A}{2}\right)\cos\left(\frac{9A-7A}{2}\right)$

$= 2\sin\left(\frac{16A}{2}\right)\cos\left(\frac{2A}{2}\right)$

$= 2\sin 8A \cos A$

(ii) $\sin 7\theta - \sin 4\theta = 2\cos\left(\frac{7\theta+4\theta}{2}\right)\sin\left(\frac{7\theta-4\theta}{2}\right)$

$= 2\cos\left(\frac{11\theta}{2}\right)\sin\left(\frac{3\theta}{2}\right)$

(iii) $\cos 8A + \cos 12A$

$= 2\cos\left(\frac{8A+12A}{2}\right)\cos\left(\frac{8A-12A}{2}\right)$

$= 2\cos\left(\frac{20A}{2}\right)\cos\left(\frac{-4A}{2}\right)$

$= 2\cos 10A \cos(-2A)$

$= 2\cos 10A \cos 2A$

(iv) $\cos 4\alpha - \cos 8\alpha$

$= -2\sin\left(\frac{4\alpha+8\alpha}{2}\right)\sin\left(\frac{4\alpha-8\alpha}{2}\right)$

$= 2\sin\left(\frac{12\alpha}{2}\right)\sin\left(\frac{4\alpha}{2}\right)$

$= 2\sin 6\alpha \sin 2\alpha$

(v) $\cos 20^\circ - \cos 30^\circ$

$= -2\sin\left(\frac{20^\circ+30^\circ}{2}\right)\sin\left(\frac{20^\circ-30^\circ}{2}\right)$

$= 2\sin\left(\frac{50^\circ}{2}\right)\sin\left(\frac{10^\circ}{2}\right)$

$= 2\sin 25^\circ \sin 5^\circ$

(vi) $\cos 75^\circ + \cos 45^\circ$

$= 2\cos\left(\frac{75^\circ+45^\circ}{2}\right)\cos\left(\frac{75^\circ-45^\circ}{2}\right)$

$= 2\cos\left(\frac{120^\circ}{2}\right)\cos\left(\frac{30^\circ}{2}\right)$

$= 2\cos 60^\circ \cos 15^\circ$

$= 2\times\frac{1}{2}\cos 15^\circ = \cos 15^\circ$

(vii) $\cos 55^\circ + \sin 55^\circ$

$= \cos 55^\circ + \cos(90^\circ - 55^\circ)$

$= \cos 55^\circ + \cos 35^\circ$

$= 2\cos\left(\frac{55^\circ+35^\circ}{2}\right)\cos\left(\frac{55^\circ-35^\circ}{2}\right)$

$= 2\cos 45^\circ \cos 10^\circ = 2\left(\frac{1}{\sqrt{2}}\right)\cdot\cos 10^\circ$

$= \sqrt{2}\cos 10^\circ$

Example 4.19

If three angles A, B and C are in arithmetic progression, Prove that

$$\cot B = \frac{\sin A - \sin C}{\cos C - \cos A}$$

Solution

$$\frac{\sin A - \sin C}{\cos C - \cos A} = \frac{2\sin\left(\frac{A-C}{2}\right)\cos\left(\frac{A+C}{2}\right)}{2\sin\left(\frac{A+C}{2}\right)\sin\left(\frac{A-C}{2}\right)}$$

$$= \frac{\cos\left(\frac{A+C}{2}\right)}{\sin\left(\frac{A+C}{2}\right)} = \cot\left(\frac{A+C}{2}\right) = \cot B$$

(since A, B, C are in $A.P.$, $B = \frac{A+C}{2}$)

Example 4.20

Prove that $\frac{\sin 5x - 2\sin 3x + \sin x}{\cos 5x - \cos x} = \tan x$

Solution

$$\frac{\sin 5x - 2\sin 3x + \sin x}{\cos 5x - \cos x} = \frac{\sin 5x + \sin x - 2\sin 3x}{\cos 5x - \cos x}$$

$$= \frac{2\sin 3x \cos 2x - 2\sin 3x}{\cos 5x - \cos x}$$

$$= \frac{2\sin 3x(\cos 2x - 1)}{-2\sin 3x \sin 2x} = \frac{1-\cos 2x}{\sin 2x}$$

$$= \frac{2\sin^2 x}{2\sin x \cos x} = \tan x$$

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